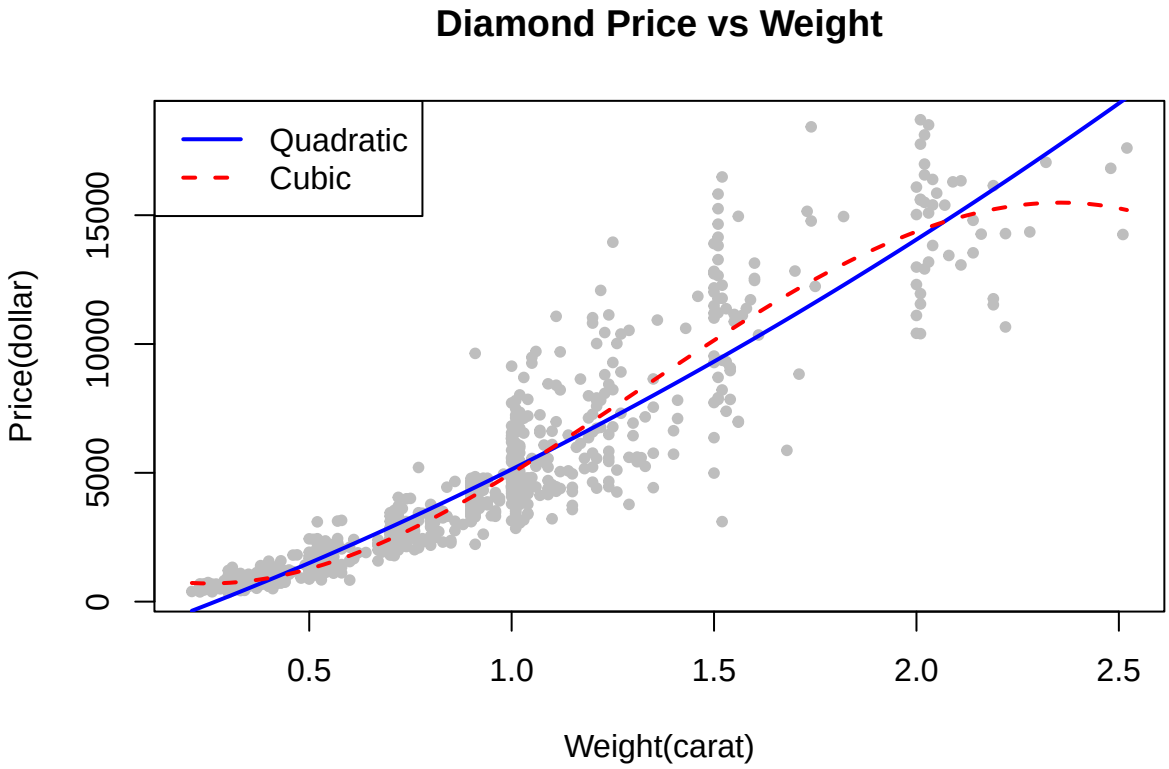
st231 lab report 1

Jingyu Zhang

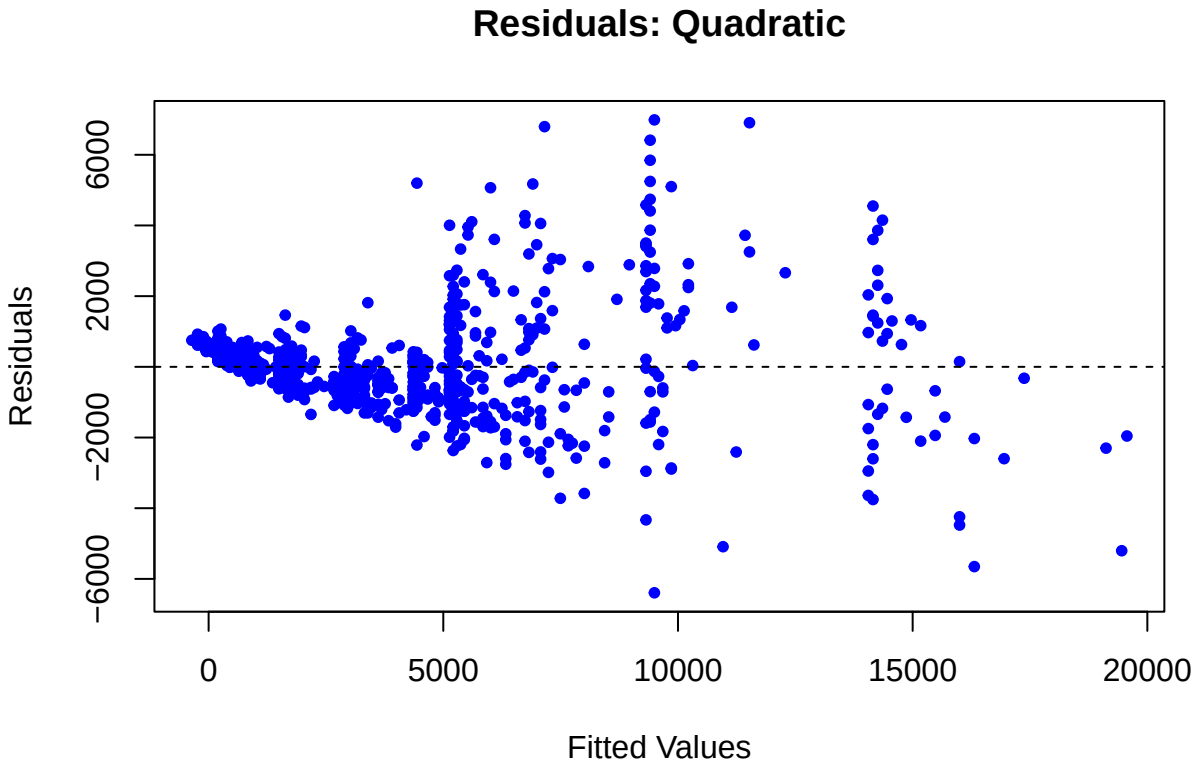
2026-01-30

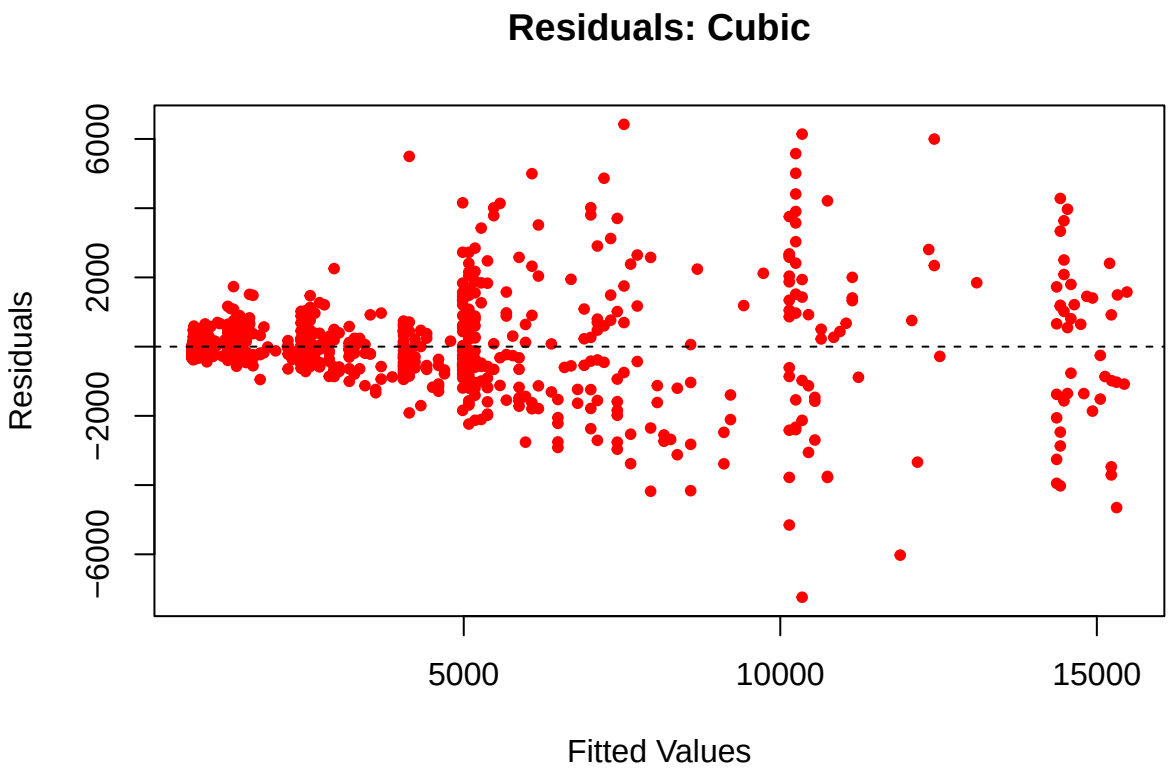
1.



2.From the scatter plot above, it is evident that both models capture the positive non-linear relationship between weight and price. However, the cubic model appears to be a superior fit. While the quadratic model underestimates when weight<0.5 carats and overestimates when weight>2.0 carats.

3.


Residuals: Quadratic
Residuals
6000
2000
−2000
−6000
0
5000
10000
15000
20000
Fitted Values



From the residual plots above, both models exhibit significant heteroscedasticity since they are both megaphone-shaped. In the quadratic plot, the residuals show a non-linear trend, whereas the cubic residuals appear more symmetric around the horizontal axis. Therefore, The cubic model is preferred.